Gluttonous Snake: A FPGA-Based Fully Unified Accelerator for Nonlinear Functions in LLMs

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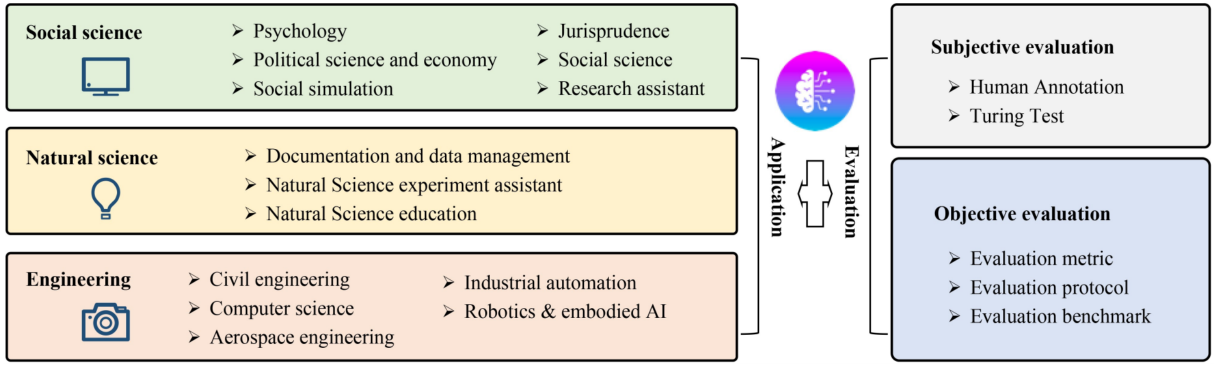
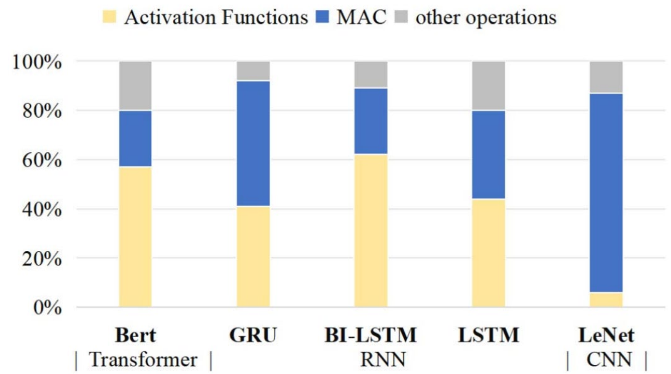
Large language models have penetrated into various fields.

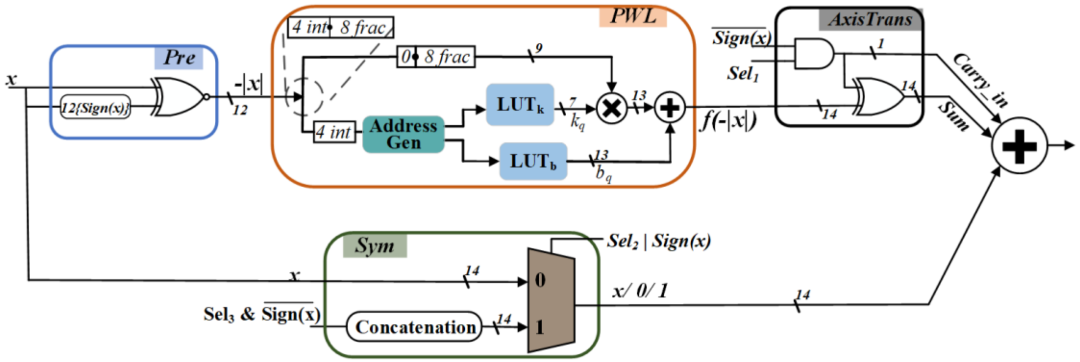
Non-linear computing is becoming a key path in the LLM reasoning process. However, there are relatively few studies related to general-purpose, low-latency, high-precision LLM non-linear operator accelerators.



Wang, L., Ma, C., Feng, X. *et al. A survey on large language model based autonomous agents. Front. Comput. Sci.* **18**, 186345 (2024).



Efficient CORDIC-based activation functions for RNN acceleration on FPGAs, ITAI, 2025.

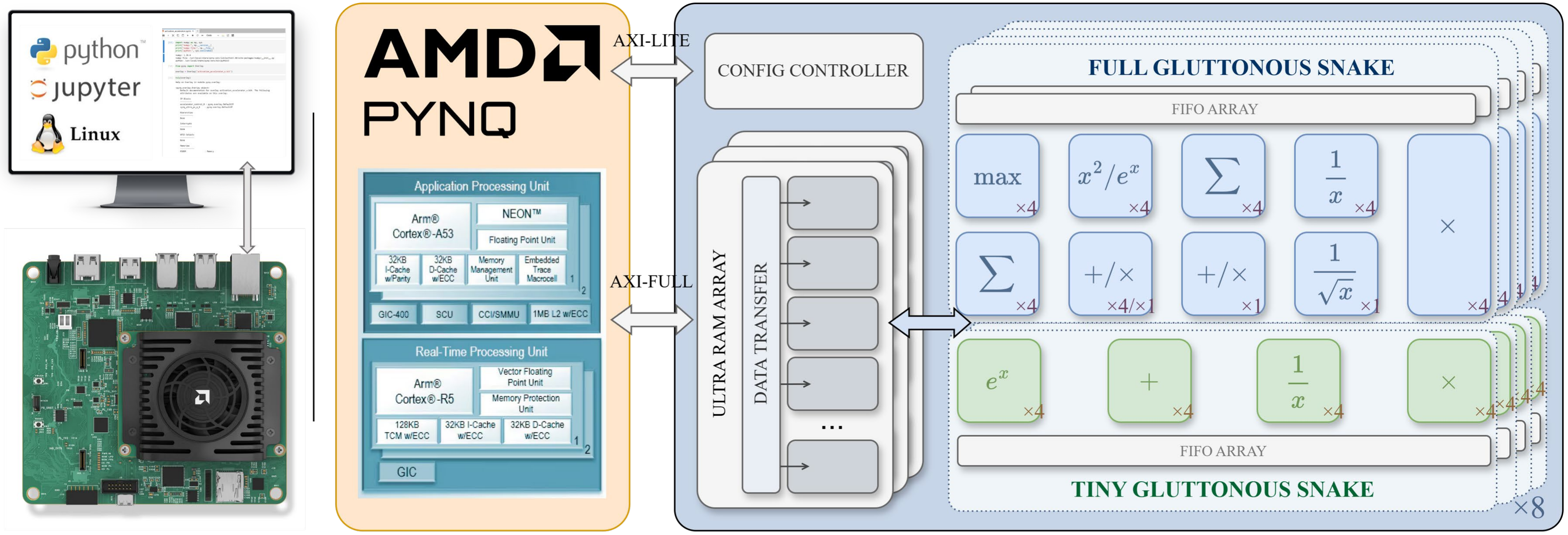


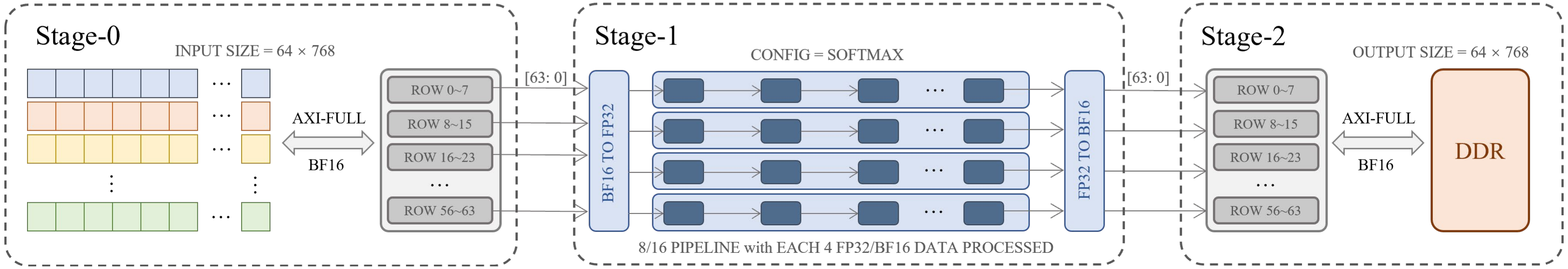
A High-Precision Flexible Symmetry-Aware Architecture for Element-Wise Activation Functions, ICFPT2021

Gluttonous Snake:

Fully-Unified, Modular, High-throughput

Non-linear FPGA Accelerator





Fully-Unified:
Support four types of nonlinear computations

Modular:
The on-chip network formed by stacking basic units

High-throughput:
Contains two "Gluttonous" blocks for fast computing

Self-Attention (SOFTMAX)

$$\frac{e^{x_{i,j}-x_{i\max}}}{\sum_{j=1}^{N} e^{x_{i,j}-x_{i\max}}}$$

Normalization (LAYERNORM / RMSNORM)

$$\frac{x-\frac{1}{N}\sum_{j=1}^{N} x_{i,j}}{\sqrt{\frac{1}{N}\sum_{j=1}^{N}(x_{i,j}-\mu_i)^2+\epsilon}} \qquad \frac{x}{\sqrt{\frac{1}{N}\sum_{j=1}^{N} x_{i,j}+\epsilon}}$$

Activation (SILU / GELU)

$$x_{i,j}\cdot\frac{1}{1+\exp(-x_{i,j})} \qquad x_{i,j}\cdot\frac{1}{1+\exp(1.702\cdot x_{i,j})}$$

Element-Wise Calaulation (E-ADD / E-MUL)

$$\mathrm{Add}(x,y)_{i,j}=x_{i,j}+y_{i,j} \qquad \mathrm{Mul}(x,y)_{i,j}=x_{i,j}\times y_{i,j}$$

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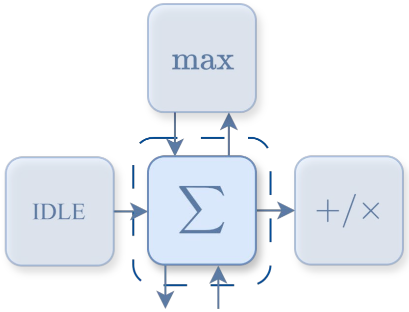
Highlight 1: **Fully-Unified**

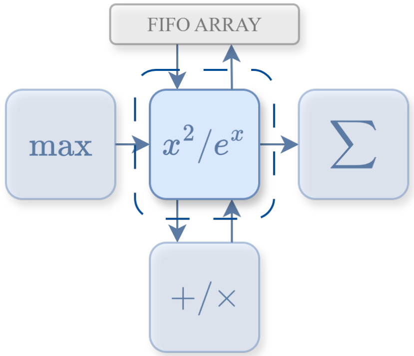
All four types of nonlinear computations are implemented on a unified architecture, based on selection register controller.

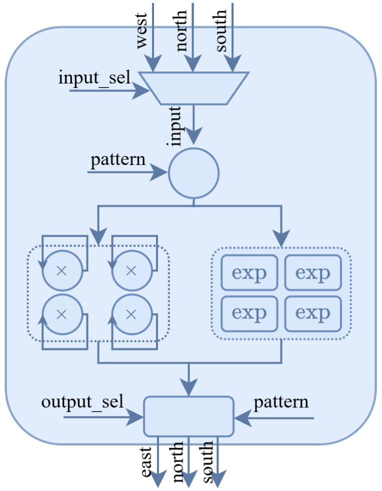
Basic Block



block_en	input_sel	output_sel
LayerNorm: Used		
1'b1 Enable	3'b010 From North	3'b001 To East
Others: By Pass		
1'b0 Unable	3'b010 From North	3'b000 Not Use

Special Block





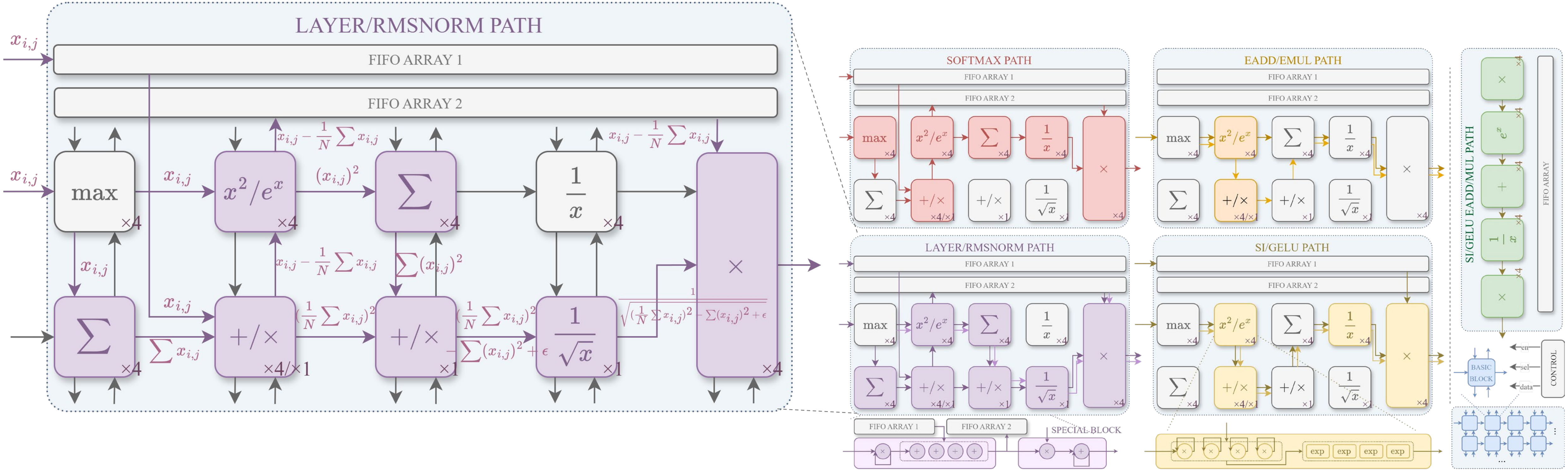
block_en	pattern	input_sel	output_sel
cal_data			
Softmax: Exp			
2'b01 Normal	1'b1 Exp	3'b100 From South	3'b001 To East
32'h00000000 Not Use			
LayerNorm/RMSNorm: Square			
2'b01 Normal	1'b0 Square	3'b001 From West	3'b001 To East
32'h00000000 Not Use			
SiLu/GeLu: Self-Cal			
2'b10 Self	1'b0 Not Use	3'b001 From West	3'b100 To South
32'hBF800000/32'hBFD9B2C7 -1/-1.702			
EMul: Mul			
2'b01 Normal	1'b0 Mul	3'b011 West&North	3'b001 To East
32'h00000000 Not Use			
Others: By Pass			
2'b00 Unable	1'b0 Not Use	3'b001 From West	3'b000 Not Use
32'h00000000 Not Use			

Core Components: Basic blocks (simple operations) & Special blocks (complex operations)

Unified Interface: 4 I/O ports(east, west, north, south) each

Configuration: Controlled via control registers

Design Advantage: High reusability of both block types and easy-to-change functions



Network Architecture & Configuration

- The network is formed by interconnecting the **basic/special blocks**.
- A central **controller** configures the entire network, directing data flow.
- It adapts to process different input data and performs **four types of nonlinear computations**, with results output to designated ports

Advantages Over Traditional Methods

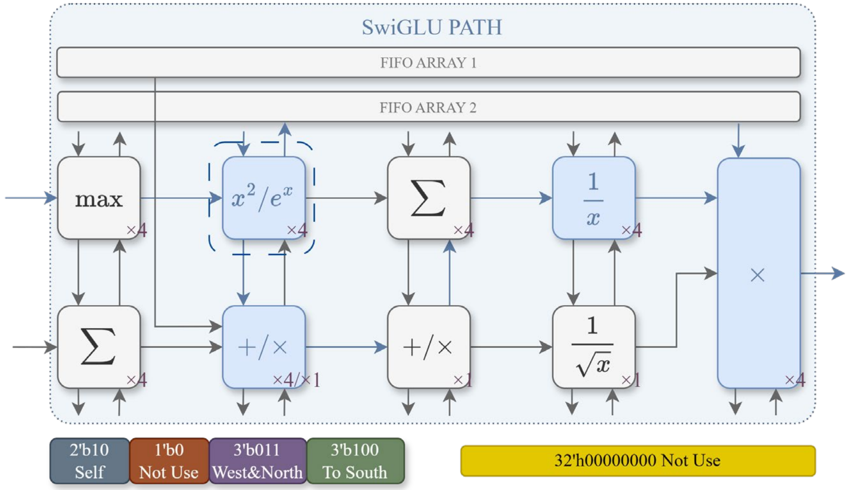
- **Simplified Implementation & Routing**
- **High Reusability & Scalability**

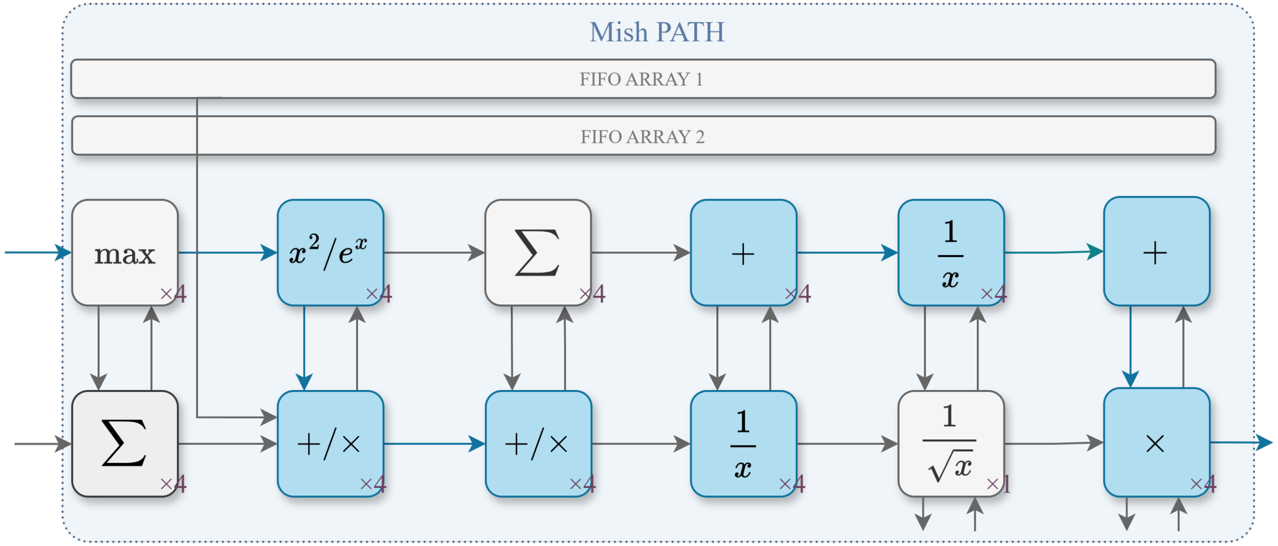
Highlight 2: **Modular**

The Gluttonous Snake can easily add or remove modules to flexibly adapt to requirements.

If we need SwiGLU: $\mathrm{SwiGLU}(x) = \mathrm{sigmoid}(x) * x$

The only modification needed is a configuration register for a block.



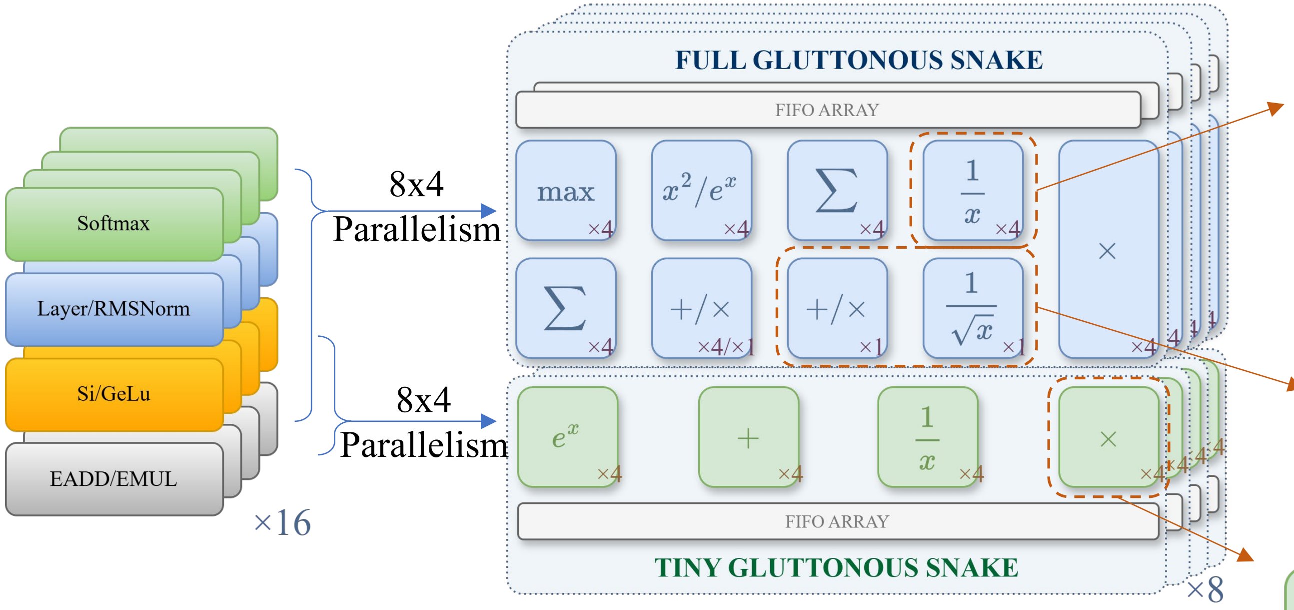


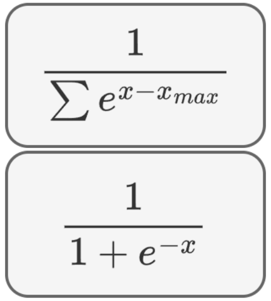
If we need Mish: $Mish(x) = x \cdot \tanh(\ln(1+e^x))$

We need to add some blocks and modify the control registers accordingly. However, this process is very simple and only requires changing a few connections of the wires.

Highlight 3: **High-throughput**

High parallelism stems from efficient resource reuse and extreme resource conservation, and is inseparably linked to precise control over the computing process and accuracy.





$$\frac{1}{\sum e^{x-x_{max}}}$$

$$\frac{1}{1+e^{-x}}$$

Since the numerical range has been restricted to a smaller range through calculation, it is completely feasible to use the FP16 computing unit.

The internal 4-channel vectors after the accumulation operator are compressed into 1 channel, so the subsequent calculations only require 1 computing unit.

Introducing BF16 multiplication and addition units to reduce resource usage.

All four types of non-linear computations can be mapped onto the FULL Gluttonous snake block, providing 8*4 parallelism. Activation non-linearity and element-wise computations can be mapped onto the TINY Gluttonous block, providing a total of 16*4 parallelism.

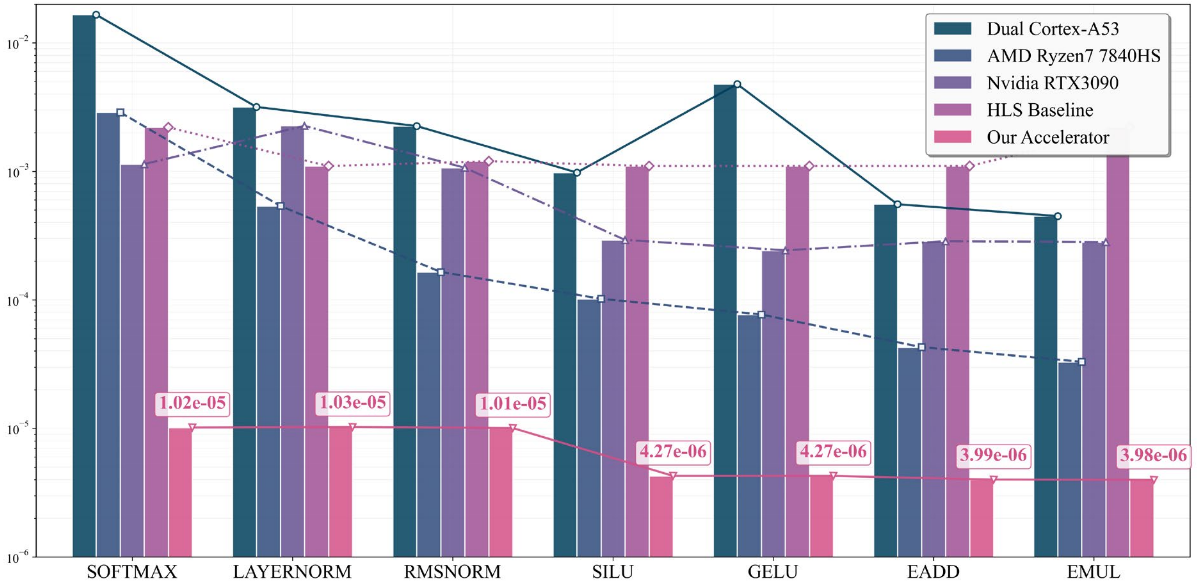
Comparison with other works

TABLE I
COMPARISON OF RESULTS

Approach	Clock(MHz)	Test Data	Delay(Cycles)	Function	Resource(LUT)
Hyft16 [1]	645	1×128	18	Softmax	19026
Hyft32 [1]	526	1×128	18	Softmax	46706
ICFPT21 [2]	410	1×1	8	SiLu/GeLu	126
Ours	300	64×768	799-2050	7 Funcs[1]	104229

[1] The measured latency in cycles for each operation is as follows: Softmax (2050), LayerNorm (1873), RMSNorm (1860), SiLU/GeLU (854), and EAdd/EMul (799).

Comparison with other platforms



The experimental results show that the greedy snake architecture effectively balances resources, accuracy and latency, and is significantly superior to general computing platforms.

Inference of Real Models

In order to better verify our acceleration performance, we ran actual models, including DistilGPT2, T5-Small and RBT3

DistilGPT2

- LayerNorm 25times
- GeLu 150times
- EAdd/Emul 25times

When I first met him...

Continuation

When I first met him in the early 1990s, he was a very nice guy. He had an amazing personality and his attitude is that you can.

T5-Small

- RMSNorm 317times
- GeLu 104times
- Emul 317times

Translate to en:
我是一个爱带着面具的句子。

Translate

I'm a love sentence with masks.

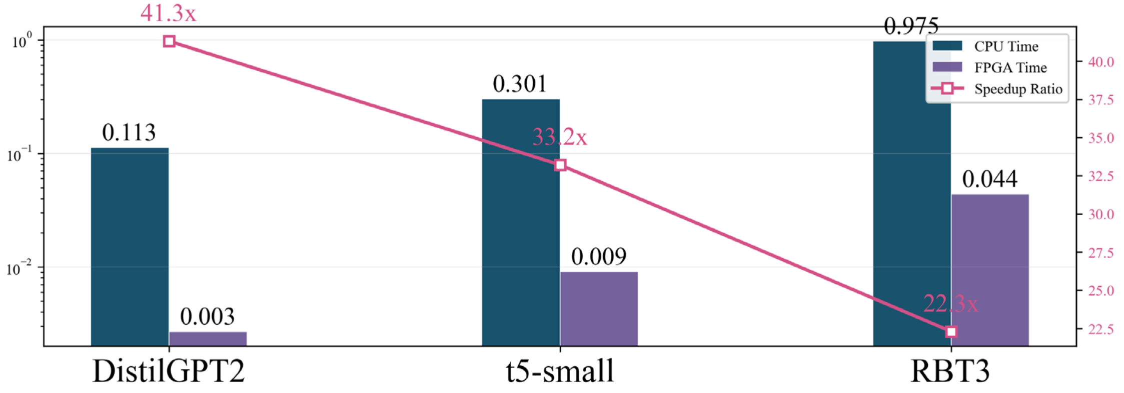
RBT3

- LayerNorm 6times
- GeLu 3times
- Emul 3462times
  Eadd 33times

我是一只[MASK]

Complete

我是一只鹿。
我是一只物。
我是一只鼠。
我是一只猫。
我是一只鸟。



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Thank You For Your Attention!

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